

Data Flow Diagram

Date	5 Jun 2026
Team ID	xxxxxx
Project Name	Smart Crop
Maximum Marks	2 Marks

Data Flow Diagram

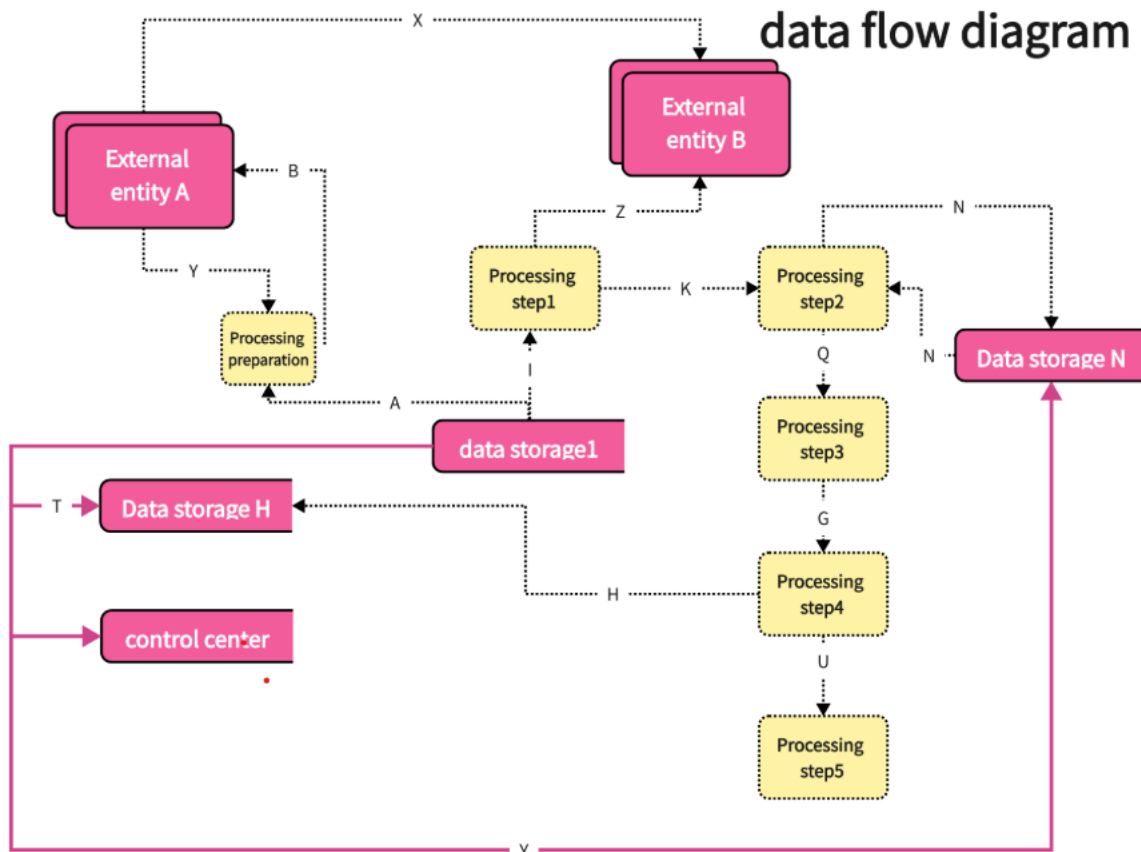
Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header Header Box [1, 2, 3]	Process	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number) Solid Rectangle	Data Store	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.



[Entity] Farmer / Researcher / Policymaker

- *Sends Soil & Climate Parameters (N, P, K, pH, temperature, humidity, rainfall) → Process 1
- *Sends Target Crop Query (Crop Type + Conditions) → Process 2
- *Sends Macroclimate Analysis Filter → Process 3
- *Receives Yield Recommendations & Analytics □ From Process 1, 2, & 3

[Process 1] Input Evaluation & Data Handling

- *Receives raw parameters from users, validates boundaries, and sends **Validated Soil/Climate
- *Metrics** → Data Store: Environmental & Crop DB
- *Triggers immediate feedback loops and logs the verification request context back to the user interface.

[Data Store] Environmental & Crop Database

*Holds primary agronomical records, optimal crop-specific thresholds, and historic regional output tables.

*Feeds ****Crop Specific Threshold Profiles**** → Process 2 (Suitability Engine)

*Feeds ****Aggregated Regional Datasets**** → Process 3 (Research Analytics Hub)

[Process 2] Predictive Suitability & Yield Optimization Core

*Processes evaluated metrics alongside threshold tables to calculate exact growth compatibility ratings.

*Sends ****Optimal Crop Recommendations & Productive Potential Metrics**** directly back to the target

****User Interface****.

[Process 3] Trend Analysis & Policy Modeling Hub

*Transforms multi-variable historical inputs into structured strategic relationship matrices.

*Outputs ****Evidence-Based Agricultural Strategies and Resource Optimization Trends**** → Policymaker UI Dashboard